



Lecture 33-EtherChannel (Port Channel)



What is EtherChannel?



EtherChannel (also called Port Channel / LAG - Link Aggregation Group) allows you to combine multiple physical interfaces into **one logical interface**.



Simple Definition:

- Multiple cables → act like **ONE cable**
- Multiple ports → behave like **ONE port**



Types of EtherChannel

◆ 1. Layer 2 EtherChannel

- Group of **switch ports**
- Works like a **single Layer 2 interface**
- Used for switching (VLANs, trunks)

◆ 2. Layer 3 EtherChannel

- Group of **routed ports**
- Works like a **single Layer 3 interface**
- You can assign an **IP address** directly



Real-Life Example

Imagine a **highway** :

- One road = slow traffic 
- Multiple lanes = faster traffic 

- 👉 EtherChannel = **multiple lanes combined as one big road**
 - 👉 More speed + no traffic jam = better performance 🚀
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⚠️ Problem Without EtherChannel

◆ Oversubscription

- End devices bandwidth > switch uplink bandwidth
- Leads to **congestion** 😞

◆ STP Issue

- If multiple links exist:
 - STP blocks all except ONE ❌
 - Only one link is used 😞

🔥 Why?

- To prevent **Layer 2 loops**
 - Otherwise → **Broadcast storm** 💣
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✅ Solution: EtherChannel

- ✓ All links are ACTIVE (no blocking)
 - ✓ STP treats them as ONE interface
 - ✓ Traffic is **load balanced**
 - ✓ No loops
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```
ASW1(config-if-range)#channel-group 1 mode ?
active      Enable LACP unconditionally
auto        Enable PAgP only if a PAgP device is detected
desirable   Enable PAgP unconditionally
on          Enable Etherchannel only
passive     Enable LACP only if a LACP device is detected

ASW1(config-if-range)#channel-group 1 mode active
Creating a port-channel interface Port-channel 1
```

EtherChannel Protocols

◆ 1. PAgP (Port Aggregation Protocol)

- Cisco proprietary 
- Max: **8 links**

Modes:

Mode	Meaning
------	---------

auto	passive
------	---------

desirable	active
-----------	--------

✓ Working:

- desirable + auto = 
- desirable + desirable = 
- auto + auto = 

◆ 2. LACP (Link Aggregation Control Protocol)

- Industry standard  (IEEE 802.3ad)
- Max: **16 links (8 active + 8 standby)**

Modes:

Mode	Meaning
------	---------

active	active
--------	--------

passive	passive
---------	---------

✓ Working:

- active + passive = 
- active + active = 
- passive + passive = 

◆ 3. Static EtherChannel

- No protocol ✗
- Manual configuration

✓ Only works:

- on + on = ✓
- on + anything else = ✗

Load Balancing in EtherChannel

◆ Based on "Flows"

- A **flow** = communication between two devices

👉 Same flow → same cable

👉 Different flows → different cables

⚠️ Why?

To avoid **packet out-of-order problem**

Load Balancing Methods

- Source MAC
 - Destination MAC
 - Source + Destination MAC
 - Source IP
 - Destination IP
 - Source + Destination IP
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Check Load Balancing

show etherchannel load-balance

Change Method

conf t

port-channel load-balance src-dst-mac

Configuration of EtherChannel

◆ PAgP Configuration

interface range g0/0 - 3
channel-group 1 mode desirable

✓ Creates:

Port-channel 1

◆ LACP Configuration

interface range g0/0 - 3
channel-group 1 mode active

◆ Static Configuration

interface range g0/0 - 3
channel-group 1 mode on

Important Rule

✓ Channel-group number:

- MUST match on same switch
 - DOES NOT need to match on other switch
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Manual Protocol Selection

```
channel-protocol lacp  
channel-group 1 mode active
```

 If mismatch → ERROR

Configure Port-Channel Interface

```
interface port-channel 1  
switchport trunk encapsulation dot1q  
switchport mode trunk
```

Running Config Example

- Physical interfaces auto-configured from port-channel

```
interface GigabitEthernet0/0  
channel-group 1 mode active
```

Important Conditions

All member interfaces **MUST** have:

- ✓ Same speed
- ✓ Same duplex
- ✓ Same VLAN
- ✓ Same trunk/access mode

 Otherwise → interface excluded

Verification Commands (VERY IMPORTANT)

✓ 1. Show Summary

show etherchannel summary

Output Meaning:

Symbol Meaning

S Layer 2

R Layer 3

U In use

P Bundled

s Suspended ✗

D Down ✗

✓ Correct:

- L2 → SU
- L3 → RU

✓ 2. Port-Channel Info

show etherchannel port-channel

✓ 3. Trunk Check

show interfaces trunk

✓ 4. IP Check

show ip interface brief

✓ 5. Running Config

```
show running-config
```

✓ 6. Load Balance

```
show etherchannel load-balance
```



Layer 3 EtherChannel Configuration

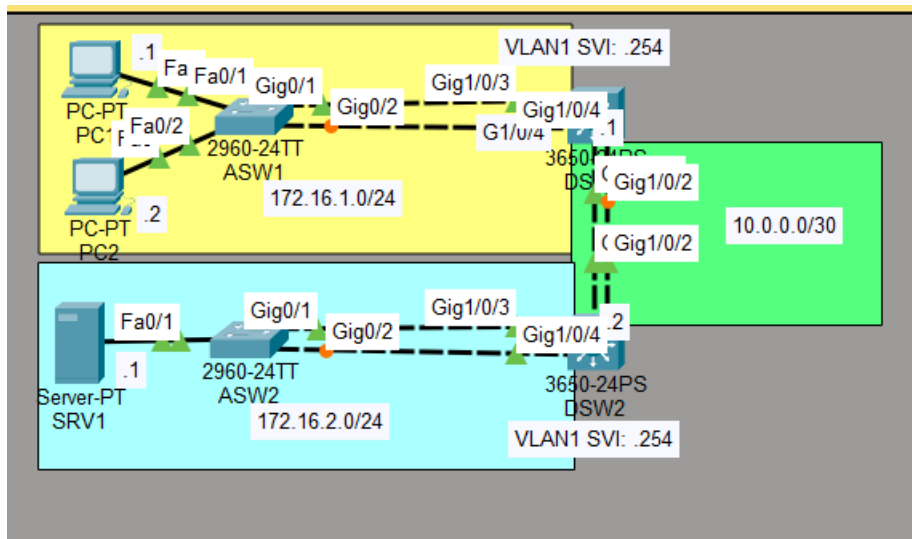
```
interface range g0/0 - 3  
no switchport  
channel-group 1 mode active
```

Then:

```
interface port-channel 1  
ip address 10.0.0.1 255.255.255.252
```

✓ IP is assigned on **Port-channel**, not physical ports

◆ Topology



- ASW1 ↔ DSW1 → LACP
- ASW2 ↔ DSW2 → PAgP
- DSW1 ↔ DSW2 → Layer 3 Static
- PCs → ASW1
- Server → ASW2

✓ 1. Configure Layer 2 EtherChannel between ASW1 and DSW1 using LACP. Configure it as a trunk.

◆ On ASW1

```
conf t
interface range g0/1 - 2
switchport mode trunk
channel-group 1 mode active
exit
```

```
interface port-channel 1
switchport mode trunk
do wr
```

◆ On DSW1

```
conf t
interface range g1/0/3 - 4
switchport mode trunk
channel-group 1 mode passive
exit
```

```
interface port-channel 1
switchport mode trunk
do wr
```

✓ **active + passive = LACP works**

✓ 2. Configure Layer 2 EtherChannel between ASW2 and DSW2 using PAgP.

Configure it as a trunk.

◆ On ASW2

```
conf t
interface range g0/1 - 2
switchport mode trunk
channel-group 2 mode desirable
exit
```

```
interface port-channel 2
switchport mode trunk
do wr
```

◆ On DSW2

```
conf t
interface range g1/0/3 - 4
switchport mode trunk
channel-group 2 mode auto
exit
```

```
interface port-channel 2
switchport mode trunk
do wr
```

✓ desirable + auto = PAgP works

✓ 3. Configure Layer 3 EtherChannel between DSW1 and DSW2 using static EtherChannel.

◆ On DSW1

```
conf t
interface range g1/0/1 - 2
no switchport
channel-group 3 mode on
exit
```

```
interface port-channel 3
no switchport
ip address 10.0.0.1 255.255.255.252
end
```

◆ On DSW2

```
conf t
interface range g1/0/1 - 2
no switchport
channel-group 3 mode on
exit
```

```
interface port-channel 3
no switchport
ip address 10.0.0.2 255.255.255.252
end
```

✓ Static EtherChannel → on + on only works

✓ 4. Configure routes to allow the PCs to reach SRV1.

- Server network behind DSW2
- PCs behind DSW1

◆ On DSW1

```
ip route 172.168.1.0 255.255.255.0 10.0.0.2
```

◆ On DSW2

```
ip route 172.16.2.0 255.255.255.0 10.0.0.1
```

✓ This allows both sides to communicate

✓ 5. What is the default EtherChannel load-balancing method used on each switch?

Check on any switch:

```
show etherchannel load-balance
```

✓ Default output usually:

```
src-dst-ip
```

👉 Meaning:

Source XOR Destination IP

✓ **6. Configure the switches to load-balance based on source and destination IP addresses.**

◆ **On ALL switches**

```
conf t
port-channel load-balance src-dst-ip
end
```

Verify:

```
show etherchannel load-balance
```

🔍 **Verification Commands (VERY IMPORTANT)**

```
show etherchannel summary
show etherchannel port-channel
show interfaces trunk
show ip interface brief
show spanning-tree
```

Output flags:

- **SU** → Layer 2 + in use ✓
- **RU** → Layer 3 + in use ✓
- **P** → bundled port ✓
- **s** → suspended ✗

⚠ **Common Mistakes**

- ✗ VLAN mismatch
 - ✗ Trunk/Access mismatch
 - ✗ Speed/Duplex mismatch
 - ✗ Wrong mode combination
 - ✗ Using "on" with LACP/PAgP
-

Quick Revision Table

Type	Protocol	Modes	Works
LACP	active/passive	active + passive	✓
PAgP	desirable/auto	desirable + auto	✓
Static	on	on + on	✓

Final Summary

- ✓ EtherChannel = Multiple links → One logical link
 - ✓ Prevents STP blocking
 - ✓ Improves speed + redundancy 
 - ✓ Load balancing using flows
 - ✓ LACP (standard) > PAgP (Cisco only)
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